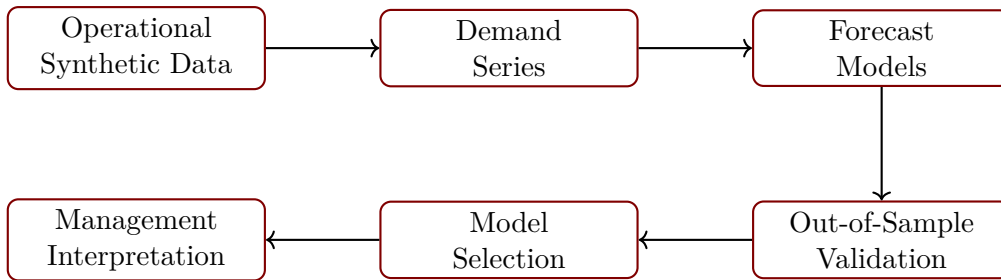


REEM V2

Demand Forecasting and Model Validation

Pharmacy Supply Chain Intelligence
Research and Development Framework

September 9, 2026



Abstract

This report presents the first demand forecasting engine developed within REEM V2. The engine is implemented entirely in pure Python without third-party Python libraries. Four transparent forecasting methods are evaluated for each medicine using an 80/20 temporal train-validation framework. Model performance is assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and forecast bias.

The underlying REEM V2 dataset is synthetic development data generated by the IDEALBUILD environment. Therefore, the results presented here are for model development and methodological validation only and must not be interpreted as operational pharmacy recommendations.

1 Purpose

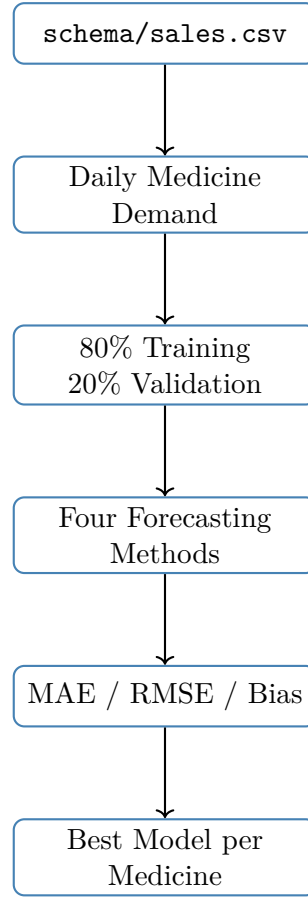
The purpose of this model is to establish a transparent baseline for pharmacy demand forecasting. The principal question is:

Given the observed demand history of a medicine, which simple forecasting method provides the most accurate prediction of future demand?

The model is deliberately transparent. Rather than assuming that one forecasting method is universally superior, REEM V2 evaluates several methods and selects the best-performing method separately for each medicine.

2 Data Architecture

The model uses the longitudinal synthetic pharmacy environment generated by `idealbuild.py`.



The current development environment contains:

- 50 medicines
- approximately 3.7 years of longitudinal observations
- 47,383 synthetic sales transactions
- four forecasting methods
- 200 model evaluations
- one selected forecasting model for each of the 50 medicines

3 Forecasting Methods

Four baseline methods are currently implemented.

3.1 Naive Forecast

The next observation is assumed to equal the most recent observation:

$$\hat{D}_{t+1} = D_t$$

This provides an important benchmark.

3.2 Moving Average

For a seven-day window:

$$\hat{D}_{t+1} = \frac{1}{7} \sum_{i=0}^6 D_{t-i}$$

The method smooths short-term fluctuations.

3.3 Weighted Moving Average

Recent observations receive greater weight:

$$\hat{D}_{t+1} = \frac{\sum_{i=1}^k i D_i}{\sum_{i=1}^k i}$$

This allows the model to respond more strongly to recent demand.

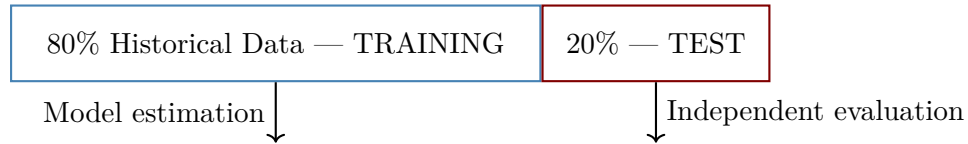
3.4 Trend Forecast

A simple least-squares linear trend is estimated over the recent observation window and extrapolated one step forward.

The trend model is intentionally simple and interpretable.

4 Validation Design

The model uses a temporal holdout rather than randomly mixing observations.



This preserves the chronological structure of the demand series.

The validation period is never used to calculate the original training forecast parameters.

5 Evaluation Metrics

5.1 Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{t=1}^n |D_t - \hat{D}_t|$$

MAE represents the average absolute forecasting error.

5.2 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (D_t - \hat{D}_t)^2}$$

RMSE gives greater influence to large forecasting errors.

5.3 Forecast Bias

$$Bias = \frac{1}{n} \sum_{t=1}^n (\hat{D}_t - D_t)$$

Positive bias indicates systematic over-forecasting, while negative bias indicates systematic under-forecasting.

6 Current Results

The model produced:

Measure	Result
Medicines evaluated	50
Forecasting methods	4
Total model evaluations	200
Final medicine forecasts	50
Validation design	80/20 temporal holdout
Validation metrics	MAE, RMSE, Bias
Python external libraries	None
Data source	SIMULATED_REEM_V2
Data status	DEVELOPMENT
Operational use	PROHIBITED

7 Example: Acetylcysteine

For medicine D0001, Acetylcysteine, the validation results were:

Method	MAE	RMSE	Bias
Naive	3.8741	5.3597	-0.0148
Moving Average	2.9698	4.1041	0.0577
Weighted Moving Average	2.9993	4.1690	0.0459
Trend	3.2256	4.4701	0.0404

The Moving Average model has the lowest MAE and RMSE among the four methods for this medicine. Therefore:

Selected model for D0001 = Moving Average

This illustrates an important REEM V2 principle:

Model selection is evidence-based and medicine-specific.

8 Analytical Results

The validation dataset contains 200 model evaluations representing four forecasting methods applied to 50 medicines. Model performance was assessed using the chronological validation period rather than the training period.

8.1 Forecast Accuracy by Method

The mean validation MAE provides a useful overall comparison of the four forecasting approaches. Lower values indicate smaller absolute forecasting errors.

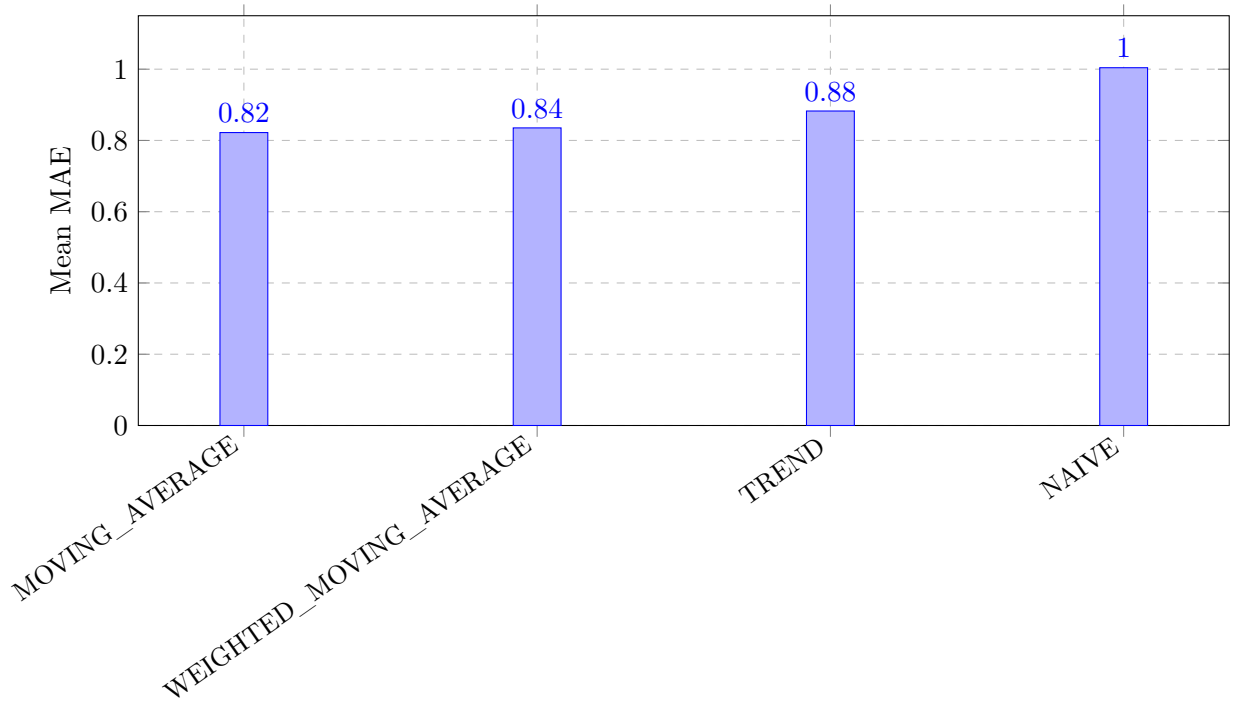


Figure 1: Mean validation MAE across forecasting methods. Lower values indicate better forecast accuracy.

The Moving Average method produced the lowest mean MAE at 0.8221, followed closely by the Weighted Moving Average at 0.8352. The Trend method recorded 0.8826, while the Naive method had the highest mean MAE at 1.0041.

This ranking indicates that, within the present synthetic development environment, recent observed demand provides a stronger forecasting signal than the simple persistence assumption or the fitted short-term trend.

8.2 Model Selection Across Medicines

The selected model is determined independently for each medicine using the lowest validation MAE.

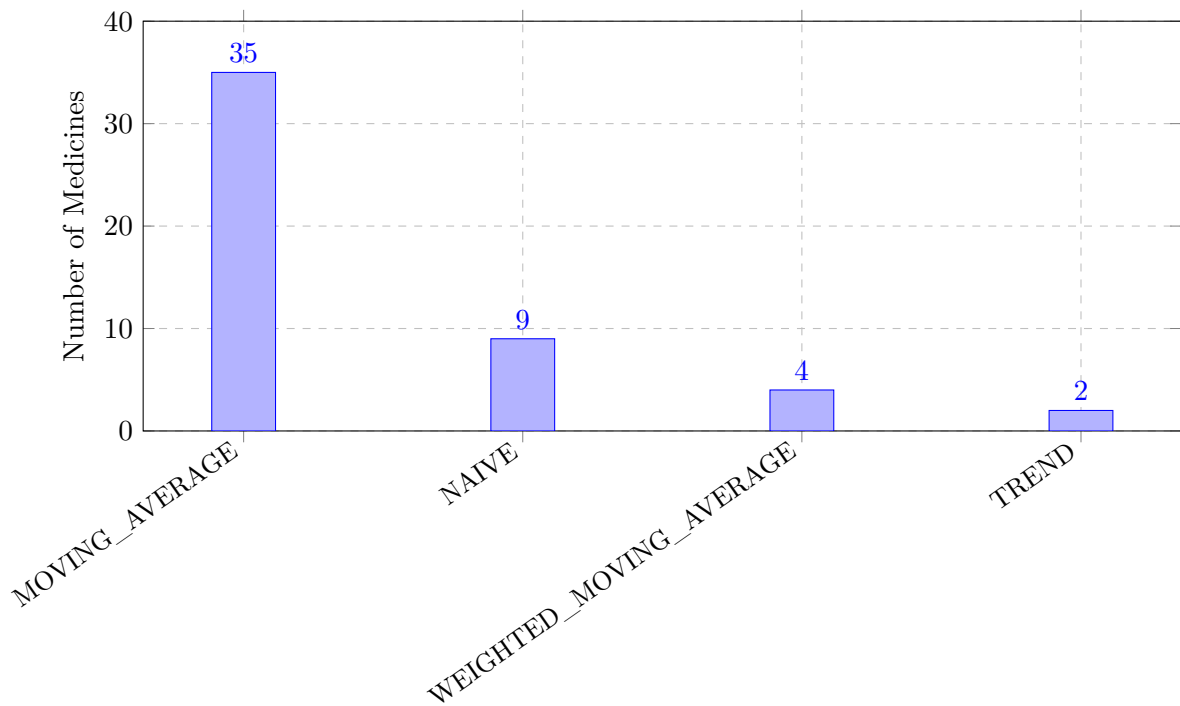


Figure 2: Forecasting model selected for each of the 50 medicines.

The Moving Average model was selected for 35 of the 50 medicines, representing 70% of the development cohort. The Naive model was selected for 9 medicines, the Weighted Moving Average for 4, and the Trend model for 2.

This result demonstrates an important REEM principle: model selection is medicine-specific. The system does not assume that one forecasting method must be optimal for every medicine.

8.3 Highest Forecast-Demand Medicines

The current selected forecasts identify medicines with the highest expected daily demand under the development model.

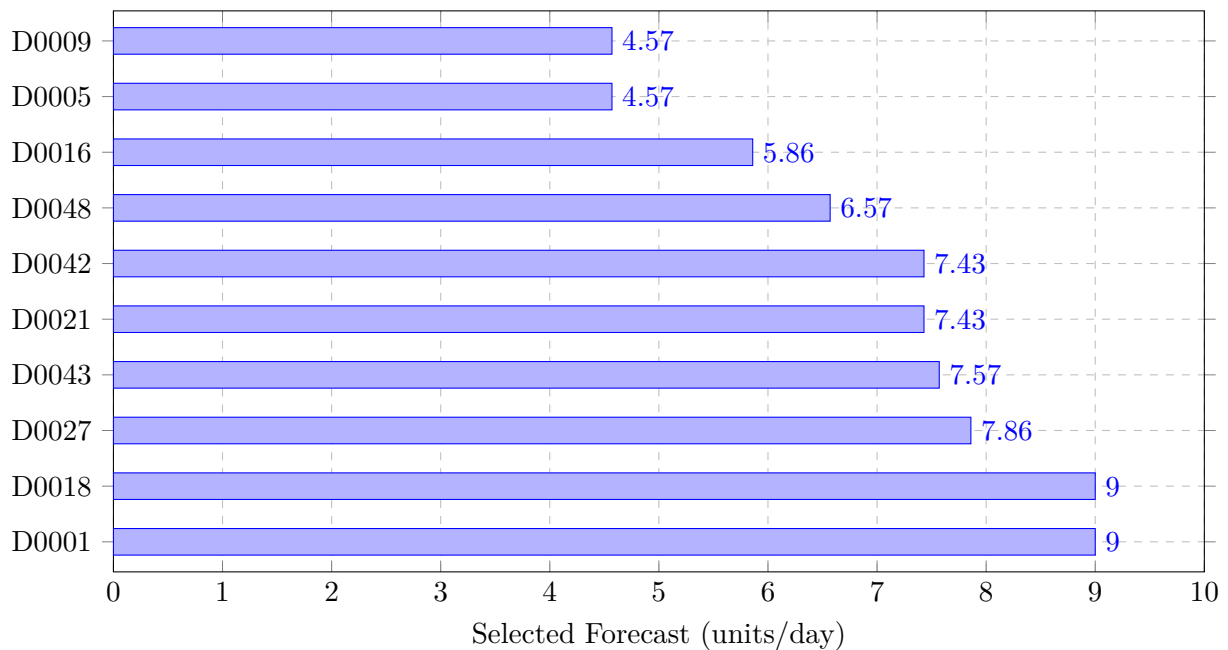


Figure 3: Top ten medicines by selected forecast demand.

The highest selected forecast is 9.00 units per day for D0001 and D0018. The remaining medicines in the top ten range from 4.57 to 7.86 units per day.

These values should be interpreted as model outputs from synthetic development data. They are not procurement recommendations or estimates of actual pharmacy demand.

8.4 Analytical Interpretation

Three findings emerge from the current experiment:

1. The Moving Average method has the lowest overall mean validation MAE among the four candidate methods.
2. Model performance is heterogeneous across medicines. The presence of four different selected models confirms that a single forecasting method is not uniformly optimal.
3. Forecast magnitude identifies medicines requiring further analytical attention, but forecast magnitude alone is insufficient for inventory decisions. Future REEM stages must integrate forecast demand with lead time, stock position, safety stock, expiry, supplier reliability, and service-level requirements.

The current result therefore represents a forecasting foundation rather than a complete inventory optimisation system.

9 Interpretation

A forecasting model should not be judged only by the numerical forecast. The quality of the forecasting process must also be considered.

For D0001, the Moving Average model achieved:

$$MAE = 2.9698$$

This means that, on average, the absolute forecast error was approximately 2.97 units during the held-out validation period.

The RMSE was:

$$RMSE = 4.1041$$

The difference between RMSE and MAE indicates that some larger errors occurred during the validation period.

The bias was close to zero:

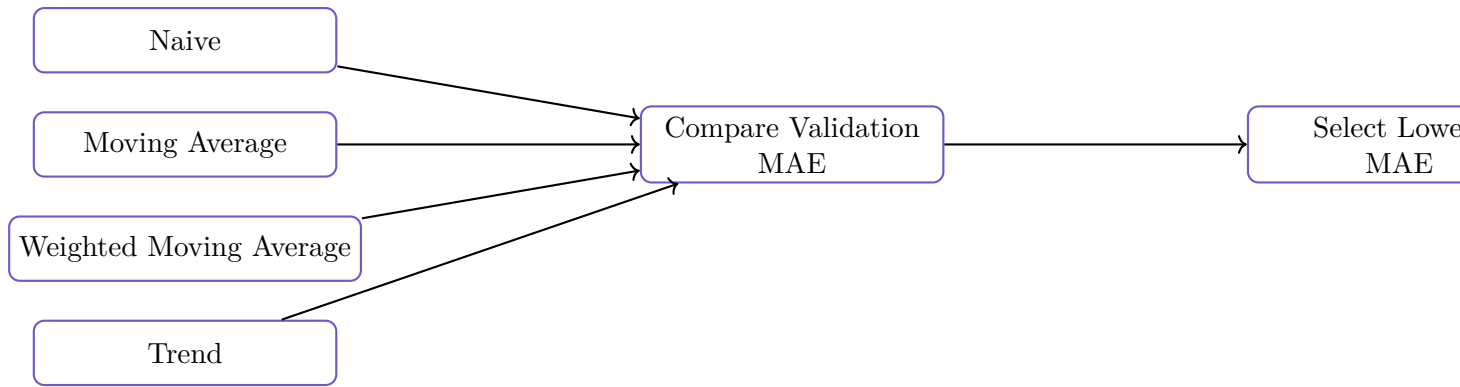
$$Bias = 0.0577$$

Therefore, there is little evidence of strong systematic over- or under-forecasting in this validation example.

10 Model Selection Logic

For every medicine, REEM V2 calculates the validation error for all four methods.

The method with the smallest MAE becomes the selected baseline model.



11 Evidence Status

The present results belong to the development stage.



The current REEM V2 demand results have reached the calculation and development-validation stages.

They have **not** reached operational trust because the underlying dataset is synthetic.

12 Limitations

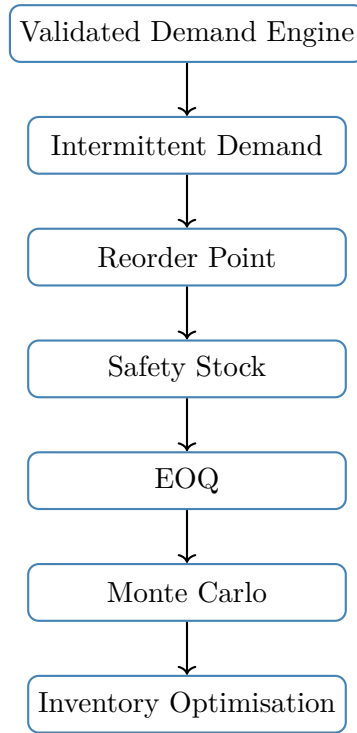
The current model has several limitations.

1. The underlying dataset is simulated.
2. The model represents baseline forecasting rather than a complete probabilistic forecasting system.
3. Intermittent demand requires dedicated methods.
4. Forecast uncertainty has not yet been fully modelled.
5. Real pharmacy data may contain stockout-induced demand censoring.
6. Calendar, promotion, substitution and external contextual variables require further development.

These limitations are intentional. REEM V2 is being developed incrementally so that each modelling layer can be inspected and validated before more complex methods are introduced.

13 Next Development Stage

The next modelling layers can build on the validated demand foundation:



14 Conclusion

The first REEM V2 demand forecasting engine successfully demonstrates a complete modelling cycle using pure Python:

Operational Structure → Demand Series → Forecast → Validation → Model Selection → Interpretation

The framework evaluates multiple forecasting methods rather than assuming that one model is universally appropriate.

The current results should be regarded as a controlled development environment for building and validating the REEM V2 intelligence architecture.

Calculated → Validated → Trusted

Synthetic development evidence must not be confused with operational pharmacy evidence.